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RESEARCH ARTICLE

Hybrid Artificial Neural Network Activation Function to Reduce Water Wastage in Agricultural Irrigation

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ABSTRACT Most nations rely heavily on agriculture to provide employment. In recent years, as a result of a growing global population's increased demand for food and water, numerous nations are consuming excessive amounts of already scarce freshwater resources. A large amount of water is used for irrigation. The purpose of this study is to find a way to reduce the amount of water lost during the irrigation process. The only practical solutions to the above problems are precision farming and smart irrigation. The study proposes a hybrid activation function based on the Artificial Neural Network algorithm. This function is used to classify the need for irrigation in various crops and predict the best time of the day for watering. The achievement of this is done by utilizing a hybrid activation function known as TanElu, which combines the Tanh and ELU functions. This function is used to classify the need for irrigation and predict the best time to water crops. The overall accuracy achieved by the proposed model was 98.24% and 97.31% in irrigation classification and prediction of the best irrigation time, respectively, which is a significant improvement compared to previous studies. Besides, execution time was 7.88 and 8.28 seconds in irrigation classification and prediction of best irrigation time respectively. The results indicate that the proposed hybrid activation function outperforms the single activation functions. Thus, the proposed model can significantly reduce water loss during the irrigation process and represents an encouraging step toward sustainable agriculture and water conservation.

INDEX TERMS Irrigation, water wastage, agriculture, artificial neural network, hybrid activation function.

I. INTRODUCTION

There are currently about 7.6 billion individuals living on Earth and the world's population is projected to reach 9.8 billion people by 2050 [1]. The biggest obstacle to sustainable agriculture and food security is the large and unbalanced population growth. The cause of the current global food crisis, which impacts approximately one billion people, is caused by an inefficient production and distribution system. If effectively reclaimed, uncultivated agricultural lands would be able to feed approximately 2-3 billion people worldwide by 2050, as stated by [2]. Water is a vital natural and indispensable economic resource for the future and sustainable development of agriculture [3]. Globally, irrigation is often scheduled based on visual inspection of crops by farmers, resulting in approximately 50% water wastage through conventional irrigation systems [4]. Controlled irrigation methods, such as sprinkler irrigation, drip irrigation, and furrow irrigation, can reduce water losses by 30-70%. However, these methods suffer from an open-loop structure, which makes it challenging to maintain accurate soil water content. Challenges presented by climate change need appropriate water management both inside and outdoors to address challenges such as population growth and the increasing

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demand for industrial and agricultural activities [5]. As a consequence, crop quality and yield are compromised, as excessive irrigation depletes soil nutrients [6]. Irrigation plays a crucial role in the agricultural industry [7]. Moreover, studies indicate that precision farming is a modern approach to agricultural management and crop production, using an array of advanced technical tools to efficiently allocate and oversee natural resources [8]. As a result, farmers need cutting-edge technology in order to effectively manage their crops using an adequate amount of irrigation water. Recent advancements in science and technology suggest that the Internet of Things (IoT) has the potential to address one of the most significant agricultural challenges, which is excessive irrigation of crops [9]. This can be accomplished by using Machine Learning (ML) algorithms and embedded system technology to automatically control the water pump. The irrigation system will be able to effectively manage water levels by utilising a collection of sensors based on the Arduino board (ESP32) [10]. The accomplishment will be achieved by measuring environmental factors, such as temperature and soil moisture, and selecting the best irrigation time. Farmers will find it easier to determine the appropriate amount of water required for efficient farm irrigation by incorporating ML techniques into the embedded system of pump controllers. This research introduces a flexible machine learning (ML) platform for the Internet of Things (IoT), where IoT and ML components communicate as services within the application context. This approach enables scalable solutions that cater to specific application requirements. The work has resulted in a platform that implements the proposed flexible architecture for IoT-ML in order to address the challenges of smart irrigation. It enables the implementation of highly customisable solutions for managing soil water. These solutions include the ability to communicate flexibly between data, physical models, and directed ML algorithms. This communication is essential for achieving important application tasks, such as estimating water needs, planning irrigation, and managing operations. This concept, known as data-driven, flexible soil water management, offers practical solutions suitable for a wide range of soils, vegetation types, and regional weather conditions. By providing farmers with smart and highly customisable irrigation solutions, this approach benefits them by reducing water and energy usage while mitigating crop yield risks through optimal maintenance of soil water content levels for plants. However, it is important to note that certain water losses, such as permeation caused by the earth, cannot be completely eliminated. Therefore, it is necessary to consider various data modeling approaches for effective water resource management. The goal is to maximise irrigation water usage while minimising water waste. The irrigation practices in Iraq, over a span of 10,000 years, have led to notable problems of waterlogging and salinity in the soils that are being irrigated. The problems have been further aggravated by the neglect of irrigated agriculture for more than two decades, resulting in a 75 percent increase in salinity levels in previously irrigated areas. Furthermore, the widespread difficulties and suffering have been caused by the absence of effective national strategies for farm water management, salinity control, and participatory irrigation management programmes. Furthermore, the effects of climate change, which encompass elevated temperatures and augmented rates of evaporation for river water (the principal source of irrigation), have played a role in the desertification of agricultural lands. Moreover, the farmers' reluctance to embrace modern irrigation methods has considerably exacerbated the situation and resulted in the squandering of significant quantities of water in agricultural irrigation. These challenges have motivated us to develop a solution aimed at regulating the irrigation process, optimizing water usage, increasing crop productivity, and enhancing overall crop quality. The goal is to create a model that facilitates the transfer of soil data to farmers through a mobile device, allowing them to receive real-time information about soil moisture levels. This information empowers farmers to make informed decisions regarding irrigation. The application developed is user-friendly and comprehensible to farmers, ensuring the maximization of water usage efficiency while reducing costs and efforts. In many countries nowadays, the main challenge is the reduction of water banks, which impacts various domains, with agriculture being the foremost affected. One critical problem is the management of intense competition for water among expanding urban centres, traditional agricultural activities, and running water uses. The balance between available and required water in the world is currently fragile. Especially after the rapid climate change, as the lack of rain and high temperatures were a major factor in the scarcity of water, even when it rains, the high temperature leads to the evaporation of a large proportion of it resulting in the drought and desertification [11]. Furthermore, the water levels in rivers and lakes are declining, and the groundwater is being depleted due to outdated irrigation systems. As a result, the soil salinity levels have increased, making it unsuitable for agriculture [12].

This study presents a contribution centered around the strategic integration of Artificial Neural Networks (ANN) with proposed hybrid activation function, known as TanElu. With a keen focus on optimizing water usage, the research encompasses several significant strides, In summary, this study's contributions are multifaceted and impactful:

- Introduction of an integrated ML-based model for sustainable water management.
- Development of an efficient model for optimal watering schedules using real-time sensor data.
- Enhancement and evaluation of the TanElu hybrid activation function to improve ANN performance in irrigation classification datasets.
- Identification of optimal watering times based on sensor readings to mitigate heat stress and water loss during critical periods.

Ultimately, this paper underscores the pivotal role of smart irrigation in driving sustainable agriculture and contributing

to global water conservation initiatives. The potential benefits for farmers, ecosystems, and water resources are substantial, reinforcing the importance of continued exploration and adoption of such innovative approaches in modern agricultural practices.

II. RELATED WORKS

A. MACHINE LEARNING IN SMART IRRIGATION

This paper developed ML models to support sustainable irrigation management by integrating ML, IoT, and web/mobile technologies for precise and continuous irrigation. They found that factors like data availability, computational complexity, and deployment style influence ML model selection. Their model outperformed the base model using supervised algorithms such as LR, DT, and SVM. Its strength lies in utilizing both supervised and unsupervised learning, including k-means and CNN, to achieve accurate results. However, the model's efficiency and performance decreased when handling larger datasets, such as environmental data, limiting its generalization ability [13].

Bhoi et al. [14] developed an IoT-enabled ML recommendation system to reduce irrigation errors by combining predicted weather data with sensor parameters. IoT devices collected environmental and terrestrial data, stored on a cloud server, while ML technologies evaluated the data to recommend irrigation. The system included an internal feedback mechanism and performed well with minimal features and data, unlike other ML algorithms requiring large datasets. However, it lacked guarantees of achieving a globally optimal solution and did not specify the amount of water needed or the best irrigation timing [14].

The authors in 2021 proposed using a linear support vector machine (SVM) to predict water intake in plants by integrating environmental factors and sensor data. The system combined moisture sensor data with SVM predictions of humidity to control irrigation based on crop water demand, soil evaporation, and transpiration. Using the MQTT protocol,predictions were transmitted to end nodes, and pump actuators distributed water accurately. The systemclassified pump valves into 10%, 25%, and 50% categories, with high accuracy observed for 10% and 25%valves. However, accuracy was lower for the 50% valve due to limited training data. While the system determined appropriate water amounts, its performance was limited by the availability of an ideal dataset [15].

The researchers proposed a distributed WSN using multiple sensors to monitor issues like excessive irrigation, soil erosion, and water waste. The system, supported by an automated ML algorithm, used Raspberry Pi 3 and Arduino Mega as control units due to their affordability and computing power. IoT and ML algorithms like SVM, SVR, and Random Forest were employed for soil classification and irrigation forecasting, with Random Forest showing the best accuracy. However, limited training data, harsh weather conditions, and non-relevance to certain research areas reduced the system's efficiency [16].

Raghuvanshi et al. in 2022 proposed an ML-based intrusion detection system to mitigate risks in IoT-enabled smart agriculture and classify intrusions in IoT networks. Data was pre-processed by converting symbolic and numeric features and classified using SVM, LR, and RF algorithms. SVM achieved the highest accuracy, outperforming RF and LR. The system's strength lies in its high accuracy compared to similar works, highlighting SVM as a reliable model for irrigation classification. However, limitations include the inability to determine optimal water levels and timing, and neglect of regional climatic conditions, reducing its generalizability [17].

This paper proposed a strategy that combines ontology and sensor data for decision-making in irrigation, with each contributing 50%. This approach integrates IoT and sensor networks, with decisions made using the KNN algorithm. Data collected an hour prior was used to forecast water requirements, and analysis was performed on an edge server before transferring results to the main IoT server and displaying them via an Android application. However, introducing an edge server increased the Internet server load, reducing response time efficiency [18].

In 2021 Glória et al. developed a sustainable irrigation system using wireless sensors, actuators, and machine learning to optimize watering strategies. The system utilized real-time and historical data via a mobile app to help users make informed irrigation decisions. Random Forest outperformed other algorithms (DT, ANN, SVM) by about 7%, achieving up to 50% water savings compared to manual methods. Combining the system with an ML strategy predicting optimal irrigation times increased water savings to 60%, conserving over 214 liters in 90 days. However, the model lacked precision for more sustainable and effective irrigation [19].

Bhavani and Thambiin 2022 proposed a decision-making system for irrigation that uses IoT sensors to collect data on temperature, humidity, and soil moisture. Classification algorithms like NB, K-NN, and SVM were employed, with K- NN outperforming the others. The system learns from historical data to determine when irrigation is needed, automatically controlling the engine to save time and labor. However, it does not account for water levels in tanks or wells, which can lead to engine damage when shutting off [20].

Integrating machine learning into smart irrigation systems optimizes water use through data-driven insights and predictive capabilities. However, these methods face limitations in classification accuracy and lack focus on predicting optimal irrigation timing. There is a need for robust algorithms that can adapt to diverse soil types, crop characteristics, and regional climates in real time. Table 1 outlines the limitations of ML classification in this context.

B. ARTIFICIAL NEURAL NETWORKS IN SMART IRRIGATION

The Authors proposed method combines recent sensor data with weather forecasts to predict precipitation using

TABLE 1. Comparison of techniques, results, and limitations in irrigation optimization.

Reference	Techniques	Results	Limitations
[14]	SVM, DT, and Naïve Bayes	SVM = 87.45% DT = 85.83% NB = 83.61%	Lack of data collected, failure to specify the suitable irrigation time, and undetermined soil type.
[16]	SVM and RF	RF = 81.6%	High cost due to wireless sensor networks (WSNs), requiring expensive battery charging and sensor compatibility.
[17]	SVM, RF, and LR	SVM = 98% RF = 85% LR = 78%	Absence of fundamental methodologies for irrigation optimization and operation.
[18]	KNN model	65%	Introducing an edge server between the IoT server and GSM module increases Internet server load, reducing response time.
[19]	SVM, NN, DT, and RF models	SVM = 79.3% NN = 80.2% DT = 77% RF = 84.6%	Upper soil depth, climatic conditions, and crop type are not well defined.
[20]	Naïve Bayes, KNN, and SVM	NB = 86.45% KNN = 91.05% SVM = 89.11%	Crop type, soil type, and suitable irrigation time are ignored.

the Ensemble learning method. By aggregating multiple models, it achieves higher accuracy and lower error rates in forecasting rainfall. The system is cost-effective due to its open-source foundation and reduced labor requirements. It automates irrigation based on soil moisture and precipitation likelihood, transmitting data to the ThingSpeak server for monitoring. However, optimizing irrigation timing and water amounts remains essential for improving efficiency, boosting crop yields, and minimizing water waste [7].

In 2022 Tace et al. proposed a cost-effective and adaptable irrigation strategy for smart agriculture using ML algorithms. Sensors (soil humidity, temperature, and rain) collected data over months, processed via Node-RED and MongoDB. Models like logistic regression, neural networks, SVM, and Naive Bayes were evaluated, with neural networks proving superior. A web application was developed for easy visualization and monitoring via phones or laptops. While the models improved irrigation control, their performance was limited by slow sampling and calibration to specific areas [21].

Rustam et al. proposed an efficient ANN-based approach to predict water quality and usage with high accuracy. The system's performance was evaluated using two datasets and utilized the Water Quality Index (WQI) as a key measure. Forecast and classification models were employed to estimate water quality and consumption, with ML algorithms such as LR, SVM, RF, DT, AdaBoost (ADA), and additive tree classifier (ET) used for analysis [22].

Taherinia et al. proposed a high-accuracy IoT system for smart greenhouse irrigation using an ANN model and data from four environment and soil sensors. A transfer learning strategy addressed challenges like limited training samples, low processing power in peripheral devices, and the integration of new sensors. The system aimed to create an adaptable and automated greenhouse irrigation solution. Data was collected every five days until an accurate model was achieved. Efficiency and accuracy could be improved by incorporating field-specific datasets and adaptively selecting sensors based on soil and crop types to reduce costs [23].

In 2021 Gu et al. proposed an ensemble neural network (NNE) technique to improve soil moisture forecasting by addressing the impact of random initial parameters on model accuracy. The NNE model independently modeled water intake and distribution, enabling daily root zone soil moisture predictions throughout the cropping season. Over 100 NN models were created with random parameters, and two methods, "single" and "average," were evaluated over three years. The system can be optimized online using soil moisture measurements and farming models. Its strength lies in accurately predicting plant water requirements but integrating climatic and temperature data could further enhance efficiency and generalization [24].

Although artificial neural networks have shown promise in the classification of smart irrigation and prediction of the ideal watering period, there are a number of restrictions that must be taken into account. Addressing these limitations will play a critical role in harnessing the full potential of ANN techniques in improving smart irrigation systems. Table 2 summarizes the limitations of Artificial Neural Networks.

TABLE 2. Limitations of artificial neural networks.

Reference	Techniques	Results	Limitations
[7]	Ensemble model (ANN, SVM, RF)	77%	Need a high computation and design time and The Ensemble learning approach did not predict rainfall time with reliable high accuracy, and this affects the irrigation process
[21]	SVM, NN,LR, and NB models	NN=97.2%, SVM=96.7%, LR=96.2%, NB=97%	The type of soil and the types of crops grown were not determined, nor was the best time for irrigation determined
[23]	ANN	A/N	The optimal amount of water for irrigation has not been determined, nor has the best time for irrigation been determined
[22]	ANN	96%	The type of soil and climatic conditions were not well defined, and the research did not address the type of crops or the appropriate time for irrigation
[24]	multilayer perceptron (MLP) neural network ensemble (NNE) method	96%	The appropriate time for irrigation, crop type and soil type are not specified

III. METHODOLOGY

The methodology employed in this paper follows a structured approach that unfolds in three key stages. The initial stage involves the comprehensive collection of relevant data. Subsequently, the process advances to the phase of constructing and refining the neural network model through enhanced algorithms, encompassing both training and evaluation. The final stage encompasses classification, prediction, and prediction script as shown in Figure 1.

A. MATERIALS AND METHOD

1) SYSTEM DESIGN AND IMPLEMENTATION

The integration of cutting-edge technologies has paved the way for sophisticated systems with unprecedented

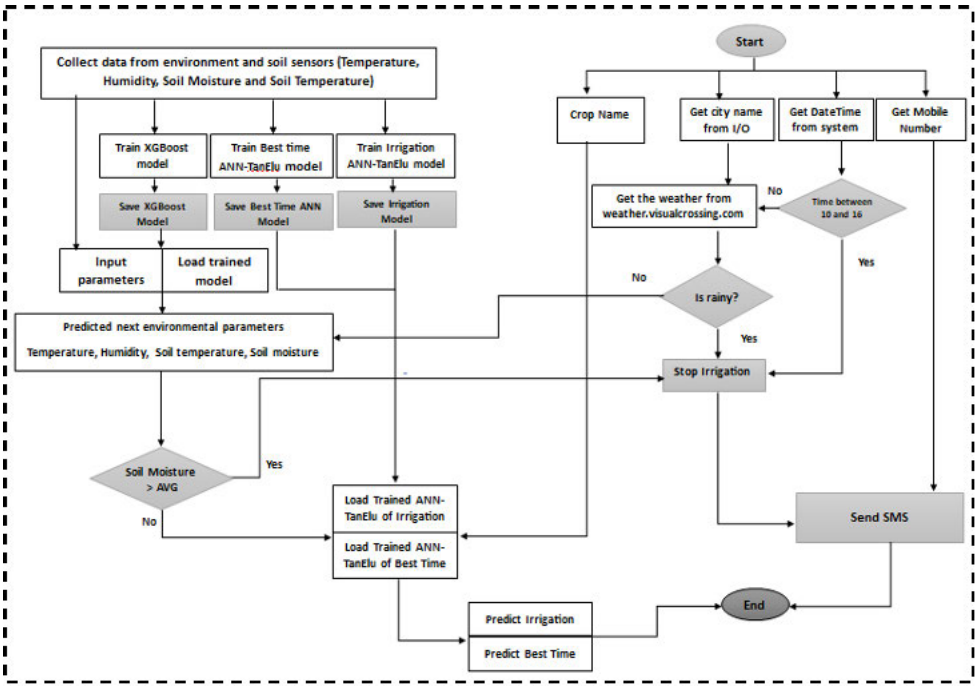


FIGURE 1. General framework of proposed model.

capabilities. The system under examination harnesses the potential of various sensors, including DHT11, PZEM-004T, Soil Moisture, and Soil Temperature sensors, to gather critical data. This data is seamlessly processed by an ESP32 microcontroller, serving as the system's neural center. The collected insights are then channeled to a dedicated card, initiating a cascade of operations. Notably, the system boasts a vigilant response mechanism, promptly notifying users through the Blynk application in the event of temperature fluctuations. Remarkably, this wealth of data is not only archived locally through an SD card data logger but is also effortlessly transmitted to a cloud platform. This cloud integration ensures real-time visualization, empowering users with mobile applications to both monitor and receive alerts. The system's LCD screen further augments its usability, displaying essential details, such as current and voltage limits, alongside the percentage of total energy consumption. Facilitating seamless connectivity, the system forges a link between temperature, humidity percentages, and Wi-Fi, facilitating data transmission to the internet. Such data availability is poised to catalyze system optimization, enabling stakeholders to access, scrutinize, and archive information, fostering informed decision-making. Beyond geographical confines, this system transcends boundaries, granting users global oversight and control through Android and iOS applications - a testament to the convergence of innovation, functionality, and convenience. While the current implementation is optimized for a controlled environment, considerations for scalability to support diverse crop types and climates, as well as adaptability to hardware-constrained rural areas, are essential. Future deployments could explore lightweight configurations and modular system designs to ensure compatibility with limited-resource settings.

2) BLOCK DIAGRAM

In the realm of innovative system design, a novel approach takes center stage, revolutionizing the irrigation process through automation and intelligent decision-making. This pioneering method seamlessly marries cutting-edge technology with practical functionality to usher in a new era of precision irrigation. At its core, the system's ingenious design orchestrates an intricate interplay of components. Its capability to issue timely alert notifications and autonomously control an auxiliary device "pump," is a testament to its sophistication. The orchestration extends further, seamlessly harnessing the power of the Internet to convey crucial notifications to the Blynk application. Figure 2 depicts a proposed IoT-based architecture design for transmitting and processing the physical parameters (soil moisture, air temperature, air relative humidity, and soil temperature) of farmland and collecting them in the cloud along with API-obtained weather forecast information.

A neural network model was developed, which consists of three separate models. The first model is used to predict the future environmental values, and it is a multi-marker ML regression model, and the other two models are deep learning models. These models are utilised to predict whether it is the best time for irrigation, and whether irrigation is necessary or not. The results are then sent to the ESP32 microcontroller unit in order to automatically operate the pump if the crop

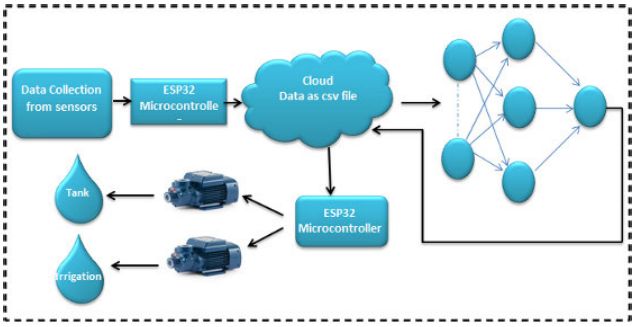


FIGURE 2. System block diagram.

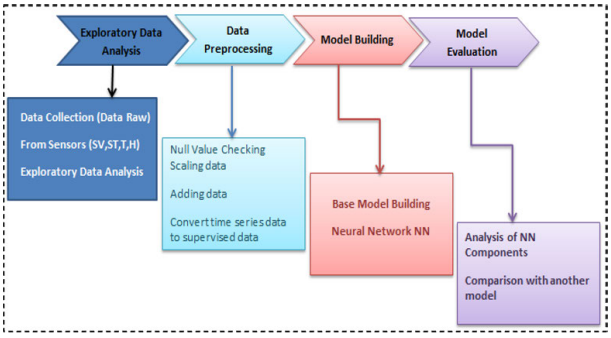


FIGURE 3. Study framework.

requires watering. The proposed activation function, which increases the efficiency of the system, is the most important factor in raising the performance of the model.

B. DATA PREPARATION

This section consists of two parts, the data Exploratory and Description, and the other is data preprocessing. Figure 3 displays the search framework.

1) EXPLORATORY DATA ANALYSIS

In our study we used two datasets

- First dataset baseline: The dataset was collected by the Departments of Computer Science and Engineering at Parala Maharaja Engineering College (Govt.), Berhampur, Odisha, India, and Government College of Engineering (Govt.), Kalahandi, Odisha, India. The Collaborative Research Scheme (CRS) of the National Project Implementation Unit (NPIU), MHRD, Government of India. The data collection period lasted for 100 days and is currently available on. https://github.com/GCEKIoTCommunity/Irrigation-Dataset. Each instance describes features or attributes of an object or entity. In the Irrigation dataset, there are 150 rows. The dataset includes seven attributes including class which indicates class/category information. The seven attributes are as follows: Crop type (1 paddy 2 barley), Crop days, and four sensors data collection characteristics (Soil Moisture, Soil Temperature, Temperature, and Humidity) are recorded. Irrigation States Yes (Y) 1, No (N) 0.
- Second dataset (dataset collected): Figure 4 depicts the manner in which the data was collected by means of our own devices over the course of one week. In this experiment, a number of features were used, including soil moisture, air temperature, air humidity, and soil moisture. These features are used as inputs. The values of the "on" or "off" in the status column acted as class labels, where "on" represents 1 and "off" represents 0. Table 3 displays the data set with 3589 rows and 7 columns. These columns include time, crop type, soil temperature, humidity, temperature, soil moisture, and irrigation. Irrigation and crop type columns are binary columns, the Time column contains the date and the time at the time of recording, and all other columns are continuous. There are only two crops in this dataset, so the crop type column contains only two values: 1 (paddy) and 2 (barley). Based on the recorded values, the irrigation column indicates whether or not irrigation is required. Therefore, this is the column that the model predicts for new reads after acquiring knowledge of the training set's patterns.

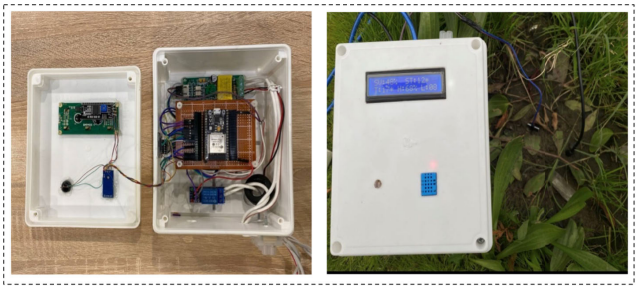


FIGURE 4. Block of sensors to collect data.

TABLE 3. Exploratory collected dataset.

Column Name	Column Type	Description
Time	Data time	Time of recording data
crop type	Categorical	1 (paddy) / 2 (barley).
soil temperature	Continuous	
soil moisture	Continuous	
Humidity	Continuous	Continuous
Temperature	Continuous	
Irrigation	need/ no need	Target column

2) DATA CLEANING AND PRE-PROCESSING

The process of "cleaning" data is required as part of the analysis process for the development of the improving model. Depending on the type of data and its primary issues, a variety of methods exist for managing this process. There are a few missing data in this dataset, and the scales of the columns vary, both of which have an effect on the performance of the model and must be resolved before proceeding.

- Handling missing value: The dataset contains both numeric and categorical data types, and each requires a distinct method for filling in missing values. Considering the absence of a few values in certain columns, simple techniques are chosen. Thus, the null value of numerical columns is filled with the meaning of each column, and for categorical data, the most frequent value of each column is chosen to fill these columns' missing data.
- Scaling Data:The primary objective of scaling is to transform data to the same range before sending them to the models for training; this can reduce the computational process and help the algorithms learn more quickly and precisely [25]. There are numerous scaling methods, including MinMaxScaler, MaxAbsScaler, RobustScaler, etc [26] which can be selected based on the data, here MinMaxScaler and Robust Scaler verified on the data that the performance of the models with MinMaxScaler was superior, so it was chosen for data scaling. This method processes the data by scaling the values to the range [0, 1], computed as:

$$X_{scale} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{1}$$

  where X is the original data; $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of each column.
- Adding new column: The farmers are aware of the optimal time for irrigation, and they choose to perform this task according to their own experience. According to the farmer's experience, the impact of sun or UV should be minimal during irrigation. This increases the effectiveness of irrigation and also is more suitable for the health of plants [27]. By incorporating the experience of farmers, we can enhance the model by utilising historical data to accurately predict the best time for irrigation. When it comes to the time of day, it is generally best to water either early in the morning or late in the evening. The reason for this is that temperatures are cooler, making it less likely for water to evaporate before it reaches the roots of the plants. In addition, watering during the middle of the day can cause water to evaporate too quickly, thereby wasting valuable resources. A new column is added to the data based on the time of previously collected data. In this new column, if the time of collection is between 8 PM and 5 AM, the value is set to 1, indicating the best time for irrigation. Moreover, Irrigation is halted between 10 am and 4 pm, as this time is considered the worst time for irrigation [28]. This is particularly true in the study area where temperatures rise during this time, thereby safeguarding crop yields from the adverse effects of high temperatures and the direct environmental impact on the plants. The column is selected as a target column and an attempt is made to predict its value based on other features such as temperature, humidity, soil moisture, and soil temperature. For any other period, the value of the column is set to 1, but a message will be sent to the farmer indicating that it is not the best time for watering, implying that it is a bad or inappropriate time for watering.

3) CONVERT TIME SERIES DATA TO SUPERVISED DATA

In order to predict irrigation, it is crucial to have future environmental information, such as temperature or humidity, forecasted beforehand. This is necessary for accurate irrigation prediction. The forecasting can be accomplished using the historical data collected by local sensors at various intervals throughout the day. The problem is a multivariate time series problem because it requires forecasting four features: temperature, humidity, soil temperature, and soil moisture. These forecasts are based on historical values of these features, which have been collected by a sensor every 3 minutes in the past. To solve this type of time series, the data should be transferred from a time series format to a supervised format to become ready for training by ML regression algorithms [29]. For this reason, the values of each column have been shifted five times and the shifted values have been saved in a new column. This is done to create t-1 to t-5 (where t represents the time interval) data for each column. These shifted values are then used as independent variables and sent to the model for training and forecasting the current value of these columns (t0) as dependent variables. After shifting, 5 new columns are added for each column of data. In each row, there are 5 previous values and the current value (from t-5 to t0). It is possible to shift data more than five times, which provides more historical data for regression model and may improve its performance, however, the number of shifting is entirely dependent on the amount of data available and the hardware performance, because doing so adds five more columns per column, increasing the complexity of the data and the computational process.

C. OVERALL VIEW OF THE METHODOLOGY (MODEL BUILDING)

This proposed work delves into the realm of smart irrigation, presenting a comprehensive approach through three separate models and the combination of them creates the final prediction. These models aim to revolutionize irrigation practices by automating decision-making processes, enhancing water efficiency, and ultimately contributing to sustainable agricultural practices. There are three phases. The first stage is Data cleaning and pre-processing, train and prediction. In Data Cleaning and pre-processing Part of the analytical process of creating an optimised model, a data ''cleaning'' procedure is necessary. There are many ways to manage this process, depending on the type of data and its main issues. At this point, the missing data is processed, and the data is scaled to the same range. The training phase which is the second stage consists of three separately models. The first model includes a solution to the time series problem. In which used to forecast the next future values of temperature,

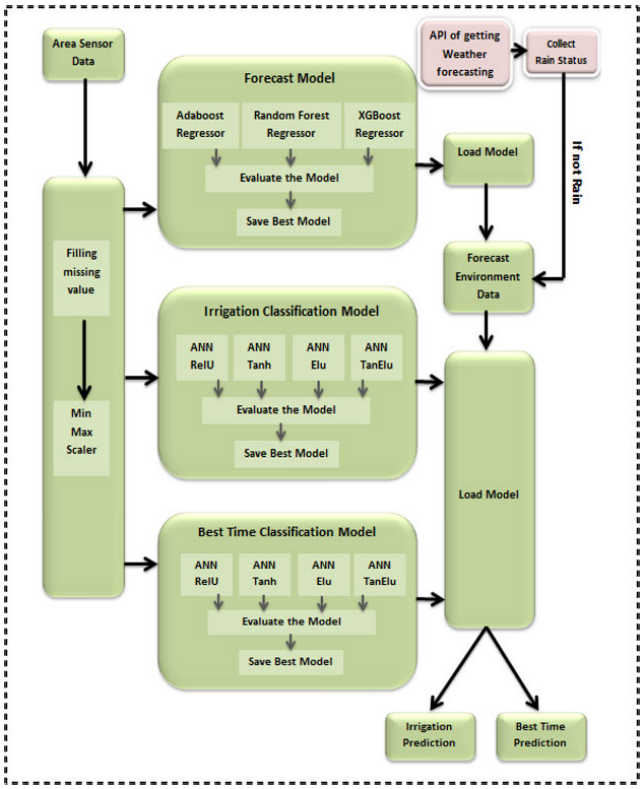


FIGURE 5. SIS using proposed activation function.

humidity, soil moisture and soil temperature. This model is a multivariate time series that is created using a multi-label ML regression model. To train a multivariate time series with ML algorithms, it is necessary to convert the time series to a supervised format. The forecasted values of this model are combined with current time and crop type and sent to the other two models. These two models were generated by the deep learning algorithm to predict whether it is a suitable time for irrigation and if irrigation is needed or not. These two models are binary classification models. In deep learning, the activation function is the most important factor in improving model performance and distinguishing between linear regression and deep learning. In this work, a custom activation function "TanElu" is proposed. After saving the best model for each trained model, it is loaded and used in the last stage to predict the future value of the environment (temperature, humidity,...) and then these values are combined with user inputs (crop type, date and time) then a message is sent to the user to inform him of the start/stop of the irrigation process and at the same time a notification is sent to the ESP32 to start/stop irrigation. Figure 5 depicts the design of the SIS utilising an ANN model that incorporates the proposed activation function.

1) FORECASTING ENVIRONMENTAL DATA

The first model is used for forecasting futures, which include area temperature and humidity, soil temperature and moisture. This is achieved by utilising previous data that has been collected from the local area through a sensor. The problem at hand is a multivariate time series problem. Due to the absence of an independent variable that can be utilised for training the model, machine learning algorithms typically cannot solve such problems. Therefore, before feeding data to models, it is vital to transform data into supervised format and makes it acceptable by regression models, this helps to create enough historical data as independent variables from current values which is crucial for building high-performance ML models. For this reason, each column has been shifted 5 times to create features and target values (20 features and 4 target columns) as explained in the preprocessing section in details. Here, three different state-of-the-art ML regression models are trained Random Forest Regressor, AdaBoost Regressor and XGBoost regressor, which are widely used ensemble learning methods in machine learning regression problems because of their enhanced predictive accuracy, capacity to manage complex interactions, and ability to reduce overfitting. After performing models, the XGBoost achieved the highest accuracy as compared to other models, thus, it is selected as the best model for forecasting. Later, to enhance accuracy, hyperparameter tuning is conducted. After experimenting with various parameters across different value ranges, the optimal parameters are determined to be a max depth of 5 and a learning rate of 0.1. The model should predict the future one step ahead. To evaluate the model, we need the last five records of data. These five records are flattened and create evaluation data. The situation is the same for forecasting the future values of temperature, humidity, soil temperature, and soil moisture. Each time, the last five historical data should be fetched and flattened from the dataset and sent to the forecasting model.

2) ANN MODELS FOR PREDICTING IRRIGATION AND BEST IRRIGATION TIME

In the field of artificial intelligence, neural networks are considered the most efficient way for solving classification problems. The model is based on the human brain and attempts to utilise the same method for discovering patterns among data. ANN, or Artificial Neural Network, utilised the capability of interconnected neurons for the purpose of data processing and the extraction of significant features [30]. The architecture of this work's ANN consists of three Dense layers with 16, 32, and 64 units, respectively. The focus of this study lies in binary classification. In the pursuit of this objective, these layers are succeeded by an additional fully connected layer boasting a single filter and employing the sigmoid activation function. This arrangement is established to compute the ultimate probability. As a countermeasure against overfitting, the models are supplemented with Dropout layers. After fine-tuning, a singular Dropout layer is integrated before the ultimate Dense layer within the irrigation model. Moreover, a single Dropout layer is added before the last Dense layer in the irrigation model. Additionally, all Dense layers in the best irrigation time

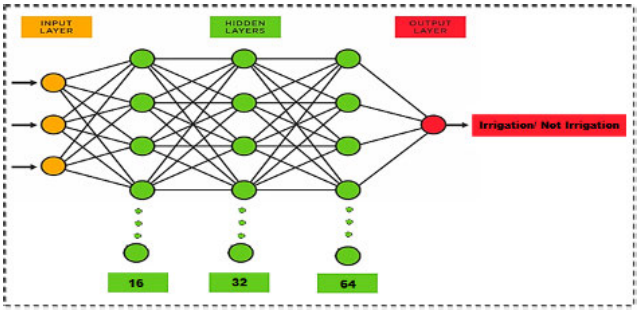


FIGURE 6. ANN architecture.

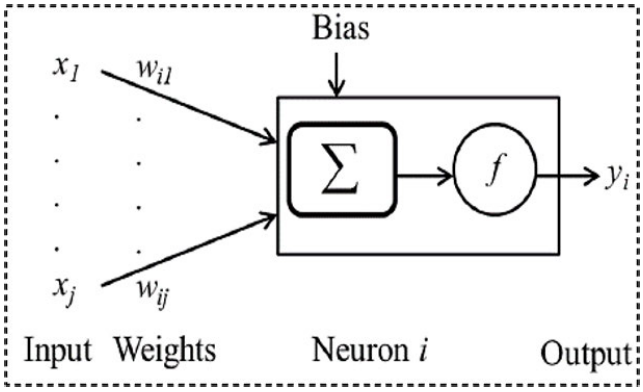


FIGURE 7. Activation function [32].

model are followed by a Dropout layer. Figure 6 shows the architecture of the ANN utilised in this study.

The input shape of ANN is (n, 10) where each row contains the 4 environmental data, the crop type and datetime of collected data which includes the year, month, day, hour, and minute. The activation function is one of the important parameters in deep learning networks. The activation function determines whether the output of a neuron in the current layer should be sent to the next layer or not. The decision is made by incorporating non-linearity into the data. Otherwise, without an activation function, a deep learning model functions similarly to linear. In this work to increase performance of ANN, a new TanElu activation function is used which is a combination of Hyperbolic Tangent Function (Tanh) and Exponential Linear Units (ELU) activation functions. Binary cross entropy is chosen as a loss function and the Adam optimizer with a learning rate of 0.01 is selected to improve the performance of the model during each epoch and the model run for 20 epochs. To evaluate the effect of the proposed activation function on the training data and compare it with the performance of other activation functions, this architecture is trained using various activation functions. The Rectified Linear Unit (ReLU), Tanh, ELU, and sigmoid are applied to the same data. Here, there are two predictions to be made; the first is whether irrigation should be performed, and the second is whether the time is appropriate for irrigation. Both predictions should be based on the same features. Thus, each feature with associated target values is sent to a distinct ANN model with the same architecture. The other process is calculating the duration of training. Before beginning training, the current time is stored in a variable, and after training, the difference between the two times is calculated to determine how many seconds training takes. This procedure applies to both ANN models.This work focuses primarily on proposing a new activation function to enhance the performance of ANN models. Different types of activation functions, their strengths and weaknesses, are described below. Finally, the proposed function, its method of calculation, and its comparative advantage are discussed.

3) ACTIVATION FUNCTIONS

The Proposed methodology involves using an ANN to classify whether irrigation is required or not. ANNs are ML algorithms that are inspired by the structure and operation of the human brain [31]. ANNs consist of interconnected nodes, known as neurons, which process and transmit data through weighted connections. The initial layer of an ANN consists of input neurons, which transmit data to the subsequent layers. The ANN is capable of comprehending more intricate objects because all of its inner layers are concealed and constructed by units that adaptively modify the information, they receive from one layer to another. The output of each node is subject to an activation function, which determines whether the neuron has enough intensity to fire or not. ReLU, ELU, Leaky ReLU, Tanh, and SoftMax. However, it is worth noting that the most commonly utilised activation functions are ReLU or sigmoid. The equation (2) follows mathematically represents a neuron:

$$y = f(w_1x_1 + w_2x_2 + \cdots + w_nx_n + b) \tag{2}$$

where Y is the neuron's output, f is its activation function, x1, x2,..., x(n) are its inputs, w1, w2,..., w(n) are its associated weights, and b is its bias to smooth out the equation's output. The inputs are multiplied by the corresponding weights for the neuron, and then the result is sent through the activation function in order to generate the output of the neuron. The activation function introduces non-linearity to the neuron, allowing ANNs to model complex non-linear relationships between inputs and outputs. Figure 7 depicts the activation function within the neuron that generates the output.

The activation functions import nonlinearity into the model and without it, the ANN works as a linear model which transforms inputs with the help of weights and bias [33]. Even with the number of hidden layers, the model cannot properly cultivate patterns without the activation function [34]. In addition, this function assists in normalising the output of neurons and enhancing the convergence of the ANN, as stated by Liew et al.(2016). Additionally, it prevents overfitting through the use of regularisation methods and dropout techniques. There are various activation functions, each with its own strength in enhancing the outcomes of the models. However, it is important to note that these functions also possess certain weaknesses. Furthermore, the

performance of each activation function is related to the data and the relationship between them. Therefore, selecting the best activation function is one of the factors to consider during hyperparameter tuning. Furthermore, activation functions provide a means to prevent dropout and overfitting, which are caused by regularisation procedures [35]. Each activation function possesses different advantages and disadvantages, and the selection of the optimal activation function relies on the nature of the problem being addressed and the desired characteristic. The activation function of ReLU is a common activation function that is mostly used to handle complex and nonlinear datasets. It was introduced by Hahnloser et al. [36] and solved the vanishing problem that appeared in the sigmoid function. However, ReLU has the disadvantage of not being differentiable at x=0 because it sets all values below 0 to zero. Sparse data may benefit from it, but if the gradient is zero, the weights cannot update, and the neurons will cease to function. The network may stop growing and operate poorly as a result of what is known as the ''dying ReLU'' phenomenon. Also, the range of ReLU is [0,) and this means it can blow up the activation [37]. Some other activation functions, such as Tanh and ELU (Exponential Linear Unit), reduce or eliminate the disadvantages of ReLU activation functions. The ELU function has a number of advantages. First, it can handle negative numbers without producing zero, which can lead to solving the ''dying ReLU'' problem. Second, it also avoids the discontinuities in the ReLU function via its continuous gradient. Finally, it has negative values that could facilitate effective learning in models by mimicking the activity of neurons in the human brain [38]. Tanh function is so similar to the sigmoid function but has an output value range between -1 and 1 same as ELU function. It can handle negative values and try to keep the output of neurons between -1 and 1 and when the output is near zero it shows the node is effective in learning more knowledge. The advantage of the Tanh function is that it is zero-centered. This can be helpful for neural network training, particularly when the input data is also centred on 0. All of these common activation functions have advantages and disadvantages, which have made researchers eager to study in this domain and propose new activation functions. In this work, it is expected to achieve higher performance by combining two activation functions that cover weaknesses of each other. The proposed activation function is created from Tanh and ELU activation functions.

4) TANELU ACTIVATION FUNCTION

This work proposes a new activation function which is called TanElu by combining the two different activation functions Tanh and ELU. The Tanh is a zero-center function that helps network to coverage faster and reduces cost of time of the training especially when the data are zero-centered too. This advantage causes it to be used mostly in hidden layers, as it keeps the meaning of hidden layers remaining near zero. On the other hand, the Tanh function output is between -1 or 1 and when the input value is large or so small, its output is be near 1 or -1 which can cause the vanishing gradient problem and in this case the neuron cannot improve more.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{3}$$

where x is the input to the activation function and e is the exponential function. where x is the input to the activation function and e is the exponential function. Equation (4) and (5) show that ELU solved this problem and it creates a more stable gradient, which in turn reduces the issue of vanishing. However, the ELU function is not zero-centred, which can lead to bias in the networks.

$$f(x) = \lambda x \quad : \text{if } x > 0 \tag{4}$$

$$f(x) = \lambda\alpha(e^x - 1) \quad : \text{if } x \leq 0 \tag{5}$$

where x is the input to the activation function and $\lambda = 1.0507$ and $\alpha = 1.6733$.

By combining Tanh and ELU, it is possible to cover the weaknesses of both. The TanElu has introduced a new nonlinearity function that helps to maintain a larger derivative for negative values and improves the learning speed. This new hybrid activation function has the benefits of both functions and overcomes limitations, resulting in a more robust and efficient activation function for deep neural networks. The TanElu activation function takes the maximum value from two individual functions, in both negative and positive space. Equation (6) and (7) demonstrate that the TanElu activation function obtains its maximum value by combining two separate functions in both the negative and positive spaces. The TanElu activation function is defined as:

$$f(x) = \max\left(a \cdot (e^x - 1), \frac{e^x - e^{-x}}{e^x + e^{-x}}\right) \quad : \forall x < 0 \tag{6}$$

$$f(x) = \max\left(x, \frac{e^x - e^{-x}}{e^x + e^{-x}}\right) \quad : \forall x \geq 0 \tag{7}$$

where:

- a is a constant that define the smoothness of the ELU.
- If x < 0 the maximum value of ELU (x) for x < 0 or Tanh(x) will select.
- If x >= 0 maximum value of x or Tanh(x) will select.

The Derivative of activation function TanElu in equation (8) and (9):

$$\frac{dy}{dx} = \max\left(a \cdot e^x, 1 - \tanh(x)^2\right) \quad : \forall x < 0 \tag{8}$$

$$\frac{dy}{dx} = 1 \quad : \forall x \geq 0 \tag{9}$$

5) PREDICTION SCRIPT

Now the models were trained and saved, and data also was prepared. To run the models a script is created, which runs the following steps one after other. At the start, it got the name of the farm location and type of crops as its input from farmers. The other data needed was time of running script

which was collected from system automatically. Based on farmer experience, 10 a.m. to 4 p.m. is the worst time for irrigation [28], [39] which reduce the quality of products and also waste water, therefore, if the time of prediction is in this period, a message is sent to the farmer and the process is halted. If the time is not within this range, the Weather API is invoked to collect information about the status of rain, and if the weather is rainy, the process is terminated, and a related message is sent to the farmer. Finally, if none of the previous criteria are applicable, the saved forecasting model is load and forecast the values of temperature, humidity, soil moisture and soil temperature of next time interval. These data are combined with crop type which enter as input by farmer and are sent to the irrigation model and best time prediction model to predict whether the farm need irrigation or not and also is it best time for irrigation or not, and the relevant message is sent to farmer based on model's predictions.

D. MODEL EVALUATION CRITERIONS

In this study, there are two different types of models which are classification and regression models; each of them has some most common metrics. For classification model, accuracy, precision, recall, and f1-score are used along the confusion matrix that shows the performance reliably. For regression model mean absolute error (MAE), mean square error (MSE) and root of mean square error (RMSE) are more general. In this work as data have some outliers, for regression models the MAE is more reliable than other metrics.

1) CONFUSION MATRIX

It is typical in literature to evaluate the model using one of the evaluation techniques, such as the confusion matrix. [40], describe the confusion matrix as a popular method for evaluating the performance of a classification model. To evaluate how well the model performs, a table is constructed to represent the number of correct and incorrect predictions about a classification problem when the actual test results are known. From the classification results, four instances can be derived. They are TP (True Positive), TN (True Negative), FN (False Negative), and FP (False Positive) [41].

- Accuracy: It is the total number of correct classifications by models. Essentially, in this work represent the number of correct irrigation and not irrigation suggestion by irrigation model and the correct number of best time and bad time predicted by irrigation time model. The equation can be expressed in this way:

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{TP} + \text{FN} + \text{FP}} \tag{10}$$

- Precision: It is a measurement standard for anticipated outcomes. It is the proportion of outcomes where the model predicts a positive sample that is really a positive sample. The formula is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

- Recall or TPR (True Positive Rate): It is a measurement standard for the original sample. It measures the proportion of samples that are expected to test positive compared to those that do. The formula is as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

- F1 score: It is a weighted average that takes into account both Precision and Recall. Only when Precision and Recall are both high and evenly distributed do F1 scores reach their peak. The equation is displayed below:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

2) MEAN SQUARED ERROR (MSE)

It is the average of square of variance between actual value and predicted value by model, it calculates through following formula:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 \tag{14}$$

where y_i denotes the actual values, $\hat{y}_i$ denotes the predicted values, and n represents the total number of samples.

3) ROOT OF MEAN SQUARED ERROR (RMSE)

It is a commonly used metric for calculating the discrepancies between values predicted by the model and observed values. The RMSE is a measure of predictive power that multiplies the magnitudes of the expected errors into a single metric. It is mainly because of its usability and simplicity in other fields. The following gives its equation:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{15}$$

4) MEAN ABSOLUTE ERROR (MAE)

It defines the average variance between the values in the dataset and the predicted values for same data by regression model.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|\text{prediction}_i - \text{actual}_i\right| \tag{16}$$

E. SOFTWARE AND HARDWARE REQUIREMENTS

Software and hardware are required for performing the experiment. This project's hardware specifications include an Intel(R) i5-4210U processor running at 2.40 GHz with 12GB of RAM and the Jupyter notebook platform. Python programming language (version 3.8), python packages such as pandas, NumPy, sklearn, seaborn, matplotlib and TensorFlow, Keras library for deep learning models, sci-kit learn and xgboost libraries for ML algorithms.

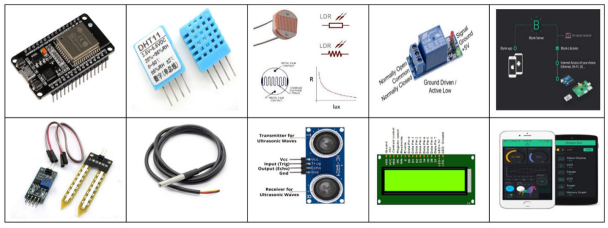


FIGURE 8. Hardware Used in Study.

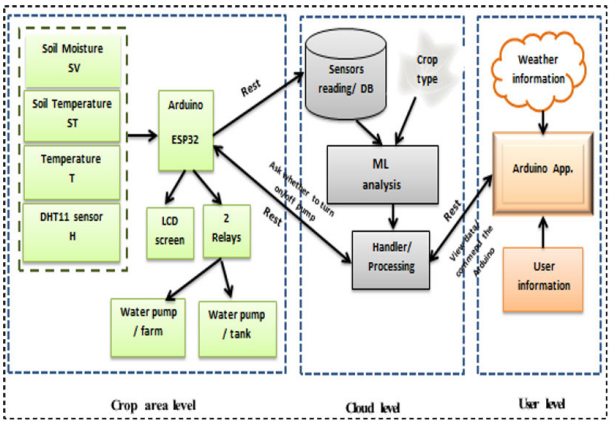


FIGURE 9. Framework for smart irrigation system based on IoT and ML.

IV. PROPOSED WORK AND COMPUTATIONAL RESULTS

This chapter discusses two main parts. The first section of this chapter presents the proposed design architecture and a comprehensive view of data collection and processing to draw conclusions, while the second section presents the computational results of experiments obtained from models with different activation functions.

A. PROPOSED DESIGN ARCHITECTURE

The architecture design and development of this project are divided into two main components hardware design and software. The hardware design and circuit planning were constructed periodically, and as a result, the project's image was engineered. While the entire, complete image was controlled by programming codes at various points during the construction of the software code. The ESP32 microcontroller, a number of sensors, and other materials are utilised intermittently in the project presented in this chapter. The LDR is used for measuring light, while the DHT11 is used for detecting weather, DS18B20 waterproof digital temperature sensor for measuring soil temperature and HC-SR04 ultrasonic sensor for measuring water tank level are examples of soil moisture sensors that measure the humidity of water content in soil. Employing a liquid LCD panel for monitoring local data makes sense. As a result, the system transmits notifications via the Blynk IoT cloud platform, which is compatible with Android and iOS devices. Figure 9 displays the different procedures required for the creation of this intelligent irrigation system.

To start, the sensors that were used to achieve the model have been selected, starting with the soil moisture sensor, which displays the soil moisture level. After that, the temperature and humidity sensors are used. Once the sensors have been connected to the ESP32 board, the board can be programmed to operate the sensors in a manner that allows for the collection and transmission of real-time data from the different sensors. The data is subsequently stored and transmitted to an ML algorithm in order to train the system to predict when the pump will start or stop. The three levels included in the frameworks, which are crop area level, cloud level, and user level, are described and explained in detail as follows:

- Crop area level: This is the first level in which various sensors are deployed in the field. Various sensors including soil moisture sensor, soil temperature sensor (DS18B20), serve as indicators to maintain the plant root in a watering condition. Additionally, a humidity sensor (DHT11), and an air temperature sensor (DHT11), are utilised to gather sufficient environmental information. All of these sensors collect data and then send it to the NodeMCU, which is an ESP32 microcontroller. The ESP32 subsequently forwards the sensors data to the cloud server. The data is collected during specific predetermined time intervals and is then forwarded to be stored in the cloud using a microcontroller. The average is calculated by collecting values and storing them as a final reading for a certain day. The HC-SR04 Ultrasonic sensor is connected to determine the water level in the tank, which is used for the irrigation process. It then transmits the data to the ESP32. The ESP32 is connected to the motor pump through the board and the relay switch, allowing for the control of the irrigation pumps on/off function.
- Cloud level: The second level is the cloud level. It is the most important level, as it is the main center and the heart of the completed system, where the process of predicting the most suitable time for irrigation takes place, and the cloud server is used to provide the service to the user. After the sensors data is stored in the database, this data along with the crop type is fed to the ML-based model to analyze and predict soil parameters, whose data were collected from the sensors placed in the field. Then, this data is fed as input to the proposed ML algorithm, where the model classifies the data into two categories that are defined as the outputs of the algorithm beforehand, which are either required (1) or not required (0). The results of these ML models are stored in the database for future procedures. The processor then, based on the classification of the ML model, sends irrigation suggestions to the user via the ESP32 application. At the same time, it sends a message to the farmer informing him of the plant's need for irrigation and turning on/off the pumps via the mobile application at the third level.
- User level: At this level, the interaction takes place between the user and the application that is downloaded

on the mobile phone, where the user's data is entered, which includes the farmer's name and login details, in addition to developing an API that connects to a weather visual site and obtains weather forecasts based on the location of the farm, if the API output shows that the weather will not be rainy, then the models will predict whether irrigation is necessary and whether or not this is the best time to irrigate. Upon receiving sensor data with an irrigation suggestion, the user may then request an on/off command to the microcontroller. If the farmer does not follow the recommendation, the feedback is sent to the server for an update.

As a component of Computer-Aided Design (CAD) system, the SMS sending API is designed to assist farmers in sending text messages to the individuals who are responsible for irrigating farms. The website used for preparing this facility was https://www.twilio.com/en-us. To receive service, one must register on the site with a valid mobile number. After registration and validation an account SID and authentication token are created as our unique identification. Farmers and irrigation professionals can receive real-time notifications of irrigation system activity, including water flow rates, valve positions, and water pressure levels, with an SMS API. This enables them to rapidly identify any issues with the system and promptly take corrective action. It also allows them to remotely schedule and control irrigation activities, which can save time, labor, and resources. The SMS API also offers the opportunity for farmers to receive weather updates and alerts, which can help them make informed decisions about irrigation scheduling.

1) EXPERIMENT SETUP

This experimental setup to implement smart irrigation based on IoT with ESP32 and an enhanced ANN model in a sandy-clay soil in Iraq. The experiment would consist of the following steps and materials (ESP32 microcontroller, Soil moisture sensors, Water level sensors, Automated water pump, Irrigation valves, Enhanced ANN model, Sandy-clay Soil). The soil type in this region is sandy-clay, as stated by [42]. This means that the fine, coarse, and medium grains are all closely situated and have a uniform size. This helps to make this soil better than sandy soils due to its water retention rate and clay soils due to their difficulty in water penetration. Due to these homogeneous proportions, the water permeability is medium. Additionally, it is characterised by moderate aeration and a good fertility rate. Consequently, dealing with it in terms of ease of cultivation and crop production is relatively easy. The Installation is following:

- Set up the ESP32 microcontroller as the central control unit for the irrigation system.
- Placing soil moisture sensors in proper areas in the soil to measure moisture content at different depths and locations.

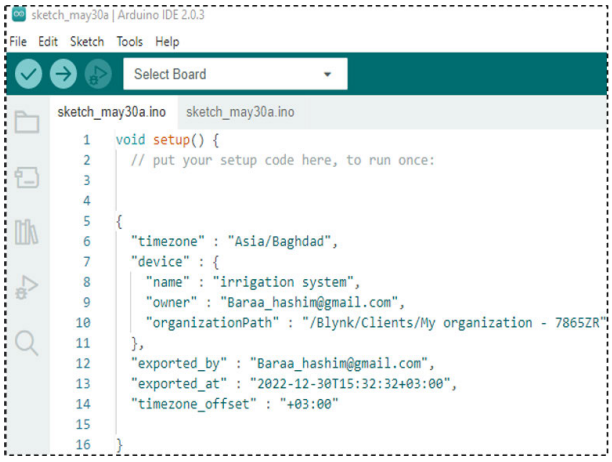


FIGURE 10. Arduino application (IDE).

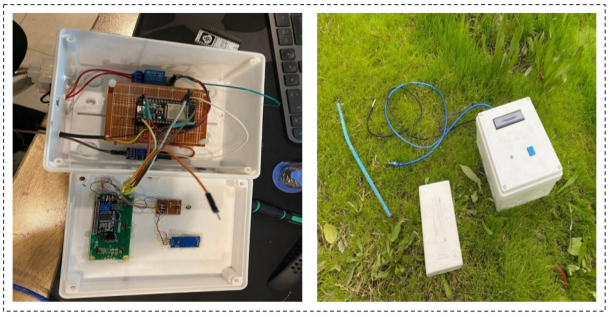


FIGURE 11. Arduino sensor testing.

- Install water level sensors to monitor water levels in the tank (HC-SR04 Ultrasonic sensor).
- Connect an automatic water pump to the water source, to supply the irrigation system with water as needed.
- Install irrigation valves to control the flow of water to different areas of the soil.
- Connect the soil moisture sensors and water level sensors to the ESP32 microcontroller using appropriate hardware and software interfaces.
- Program the ESP32 microcontroller to receive data from the sensors, process the data using the enhanced ANN model, and respond by controlling the irrigation valves and water pump to deliver appropriate amounts of water to the soil.
- Perform experiments over time to determine the effectiveness of the intelligent watering system. Make any necessary adjustments to the configuration and algorithm.
- Record and analyse data on soil moisture levels, weather patterns, and water consumption in order to optimise the system's efficiency and effectiveness.

Figure 10 is show the setup code for Arduino IDE application that connected to the attached sensors as shown in Figure 11.

2) STUDY IMPLEMENTATION

A variety of sensors are used in smart irrigation systems, including temperature sensors, humidity sensors,



FIGURE 12. Blynk application mobile App dashboard.

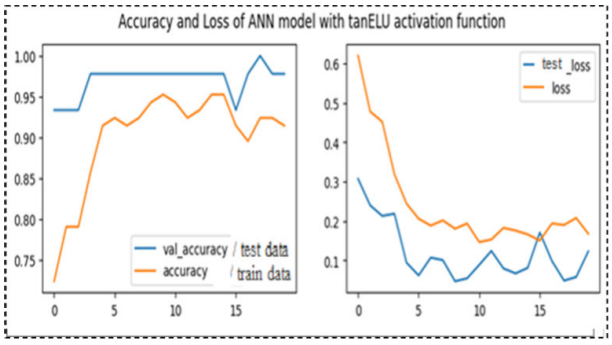


FIGURE 13. Accuracy of the proposed TanElu of ANN model.

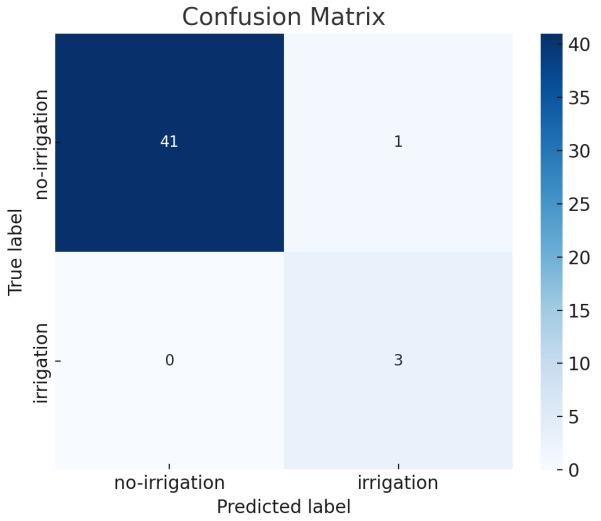


FIGURE 14. Confusion matrix of ANN model by TanElu.

light sensors, soil temperature sensors, and soil moisture sensors. However, the soil moisture sensor is particularly crucial in decision-making regarding irrigation frequency. The Blynk application dashboard seamlessly integrates with these sensors, allowing users to monitor the measured data remotely. Figure 12 displays the Blynk mobile application dashboard, which visualizes the data, making it easy for users to monitor the collected data sent to the cloud. This visualization helps indicate the appropriate time to make the irrigation decision by using the proposed classification.

B. DATASET TRAIN-TEST SPLITS

To avoid overfitting and ensure unbiased results, the dataset is divided into train and test subsets. For forecasting temperature, humidity, soil moisture, and soil temperature using an ML regression model, 90% of the data is used for training, while 10% is reserved for evaluation. This approach relies on time series data, with predictions based on the last five records transformed into a supervised format. Two additional models predict irrigation necessity and determine suitable irrigation timing. These datasets are split in an 80:20 ratio, with 80% used for training to identify patterns and 20% for testing. This range is commonly adopted when data is sufficient, ensuring robust model performance.

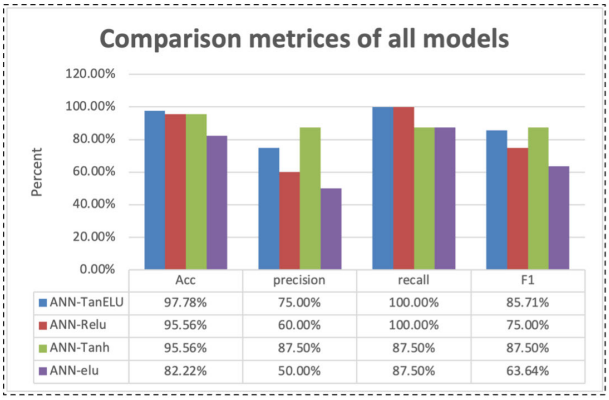
C. COMPARISON RESULTS OF EXPERIMENTS

1) RESULTS OF TANELU ACTIVATION FUNCTION FOR IRRIGATION MODEL FOR THE BASELINE DATASET

The initial dataset presented in the study was used to train the proposed model and compare it to the base model. On this data, the model can achieve 97.78% accuracy. The other metrics are 75%, 100%, and 85.71% for precision, recall and f1-score respectively. As the dataset is small the model took 2.43 seconds for training. Figure 13 shows the improvement in accuracy and loss during epochs resulting from the use of the ANN model with the proposed TanElu activation function in the paper. During the first five epochs, it is observed that the accuracy of train and test data improves, but after that, they remain nearly the same, as is often the case with small datasets. In the loss, the rate of improvement is almost identical to the accuracy, but this was only the case for the first five epochs; since then, it has fluctuated around the same value.

The rationale behind this lies in the fact that while the SVM algorithm is robust, it is usually used for classification and regression tasks using a hyper-optimal level that separates well the different classes in the data. However, their performance may stabilize when dealing with complex data sets or nonlinear relationships, while neural networks have a superior ability to capture and deal with complex nonlinear relationships in the data. The improvement in accuracy was also due to the use of the proposed hybrid activation function (TanElu) which allowed modeling of a wider range of relationships within the data. Figure 14 demonstrates that only one record was incorrectly classified, and the model performed well for the remaining data.

Figure 15 displays demonstrates that the ANN model with the TanElu activation function achieves the highest performance with 97.78% accuracy. This is because it can pick up complex patterns and relationships in data, which makes them better suited for complex classification tasks. In addition, its smoothness helps mitigate the problem of a



	Acc	precision	recall	F1
ANN-TanELU	97.78%	75.00%	100.00%	85.71%
ANN-Relu	95.56%	60.00%	100.00%	75.00%
ANN-Tanh	95.56%	87.50%	87.50%	87.50%
ANN-elu	82.22%	50.00%	87.50%	63.64%

FIGURE 15. TanElu and other activation.

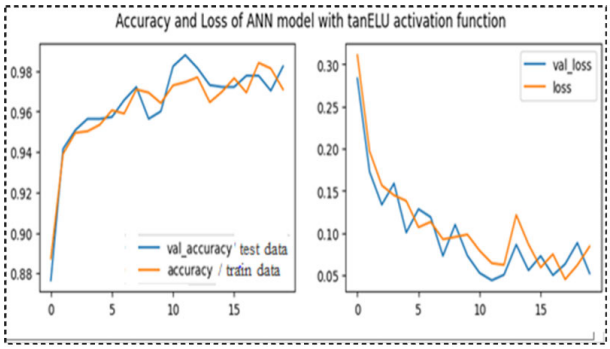


FIGURE 16. Accuracy of the proposed TanElu.

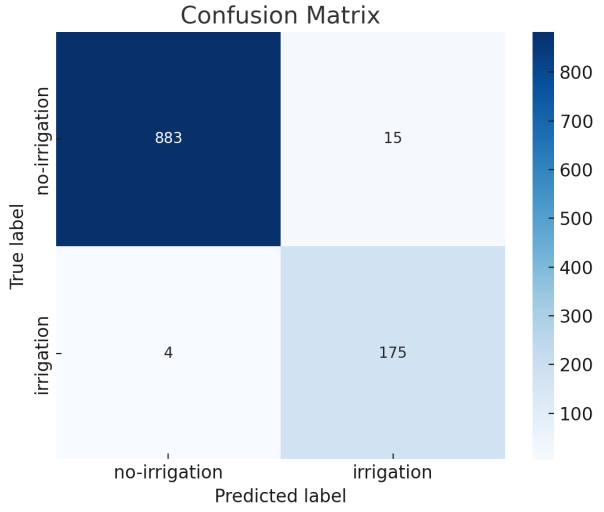


FIGURE 17. Confusion Matrix of ANN model by TanElu.

vanishing gradient, allowing for a more stable and effective workout. On the other hand, the combination of Tanh and Elu properties provides a balanced combination of non-linearity and effective gradient diffusion, which contributes to improved model accuracy, this accuracy demonstrating the efficiency of the proposed activation function. Afterwards, the Tanh and ReLU activation functions attained the same accuracy of 95.56%, whereas the precision and F1 score of 87.50% in Tanh demonstrate superior performance compared to ReLU, and the metrices demonstrate that Tanh is a more generalised model. The ELU has lowest accuracy, which is 82.22%. The precision, recall, and F1 score are 50%, 87.50%, and 63.64% respectively. There is no sign of bias towards any specific class, and the models performed well in classifying the data.

2) RESULTS OF TANELU ACTIVATION FUNCTION FOR IRRIGATION MODEL FOR THE COLLECTED DATASET

Within the context of a comprehensive dataset gathered from a local farm. This dataset presents a real-world scenario that challenges the neural network's ability to discern intricate patterns and relationships within the data. The attention-grabbing aspect of this paper lies in the remarkable achievement of the TanElu activation function, boasting an unparalleled accuracy rate of 98.24% on this dataset. This accomplishment is accompanied by commendable metrics in precision, recall, and f1-score, attaining 92.11%, 97.77%, and 94.85% respectively. The following Figure 16 shows the improvement of the accuracy and loss throughout epochs. It is observed that the accuracy of both train and test data gradually improves, although the rate of improvement was higher during the first four epochs. During the loss, the speed of improvement is greater in the first 10 epochs, and thereafter, it fluctuates slightly. The model's training took 7.88 seconds. This improvement is attributed to the use of the proposed activation function (TanElu). This combination enabled the network to capture a wide range of data patterns and relationships that might not be effectively captured by a single activation function.

Based on the confusion matrix that is shown in Figure 17 The model exhibited good performance, with only 19 records of data classified incorrectly. The number of correct classifications for both classes is nearly balanced, which indicates that the model does not exhibit bias towards any specific class.

The Figure 18 shows that the ANN model with TanElu activation function has the highest performance, achieving 98.24% accuracy. This result demonstrates the effectiveness of the proposed activation function. After that, the accuracy of the model decreased by approximately 2% for ReLU, and by approximately 3% for ELU and Tanh, respectively.

The enhancement in achieved accuracy hinges on the utilisation of the proposed hybrid activation function, Tanh-Exponential Linear Unit (TanElu), within an artificial neural network (ANN). The underlying aim of incorporating this novel hybrid activation function is to harness the combined virtues of both Tanh and ELU. The employment of Tanh lends the advantage of mapping values to a range between -1 and 1, resulting in a centered distribution around zero. This attribute greatly aids the network's convergence during the training phase. Conversely, ELU comes into

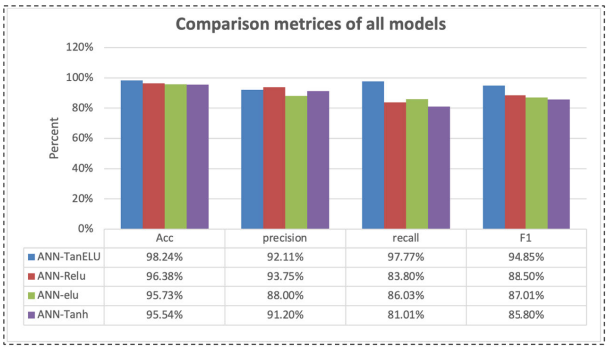


FIGURE 18. TanElu and other activation for collected dataset.

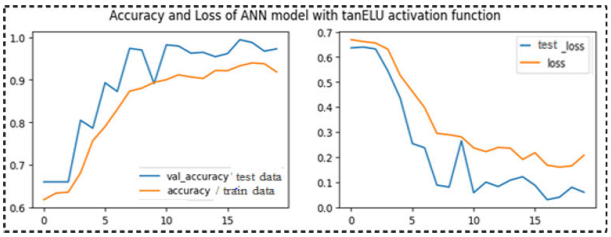


FIGURE 19. Accuracy of TanElu to predict best time for irrigation.

play to mitigate the vanishing gradient issue, facilitating expedited learning by permitting negative values—an aspect particularly advantageous for deep networks. Moreover, the dataset itself encompasses intricate, real-time nonlinear relationships among its data, imparting a heightened level of complexity. In this context, the amalgamated activation function proves more adept at capturing these multifaceted intricacies, notably in contrast to the relative simplicity of single activation functions.

3) RESULTS OF TANELU ACTIVATION FUNCTION FOR BEST TIME MODEL FOR COLLECTED DATA

Real-time environmental data may contain noise due to measurement inaccuracies or unexpected events. Some noise in the data may lead to gradients during backpropagation and thus cause fluctuations in accuracy and loss of validation data. Figure 19 below displays the proposed TanElu activation function, obtains the highest accuracy of 97.31%. While the precision, recall and f1-score of this model are 93.99%, 98.36% and 96.13% respectively. The proposed hybrid activation function, by combining the strengths of Tanh and ELU, better deals with noise and prevents it from significantly affecting predictions compared to other noise-sensitive active functions. It can be seen through a continuing trend of improvement throughout the training process. The training of this model took 8.28 seconds. The accuracy of the model is observed to gradually increase, with the improvement occurring at higher steps during the first 10 epochs. The loss value of both the train and test data decreases moderately in 10 epochs, and after that, it decreases slightly and almost remains the same.

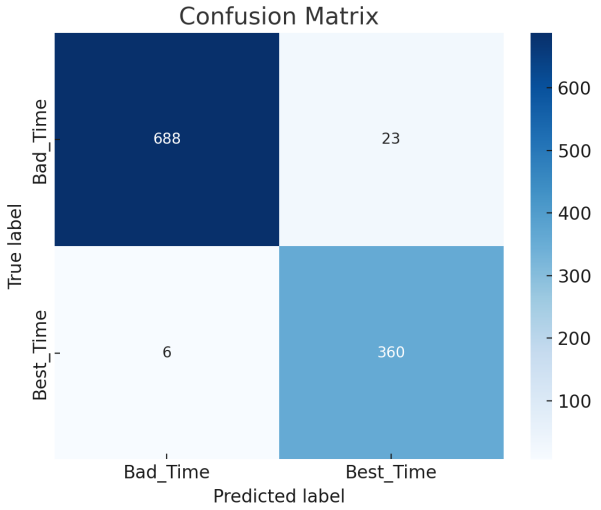


FIGURE 20. Confusion matrix of best time prediction.

In contrast, Figure 20 of confusion matrix shows that the model can correctly classify 1048 data correctly and 29 cases classify incorrectly. The confusion matrix shows the model's performance with acceptable bias and variance towards both classes.

Watering decisions need to adapt to rapidly changing conditions. The use of the ELU activation function in the Proposed Activation Function (TanElu) is important in mitigating the vanishing gradient problem, ensuring that the network can quickly adapt to new information and adjust predictions. The hybrid activation function adapts to sudden shifts in weather, soil moisture, and other variables, providing more accurate watering timing predictions in real time. In addition, the proposed activation function (TanElu) stabilizes the training dynamics, preventing rapid fluctuations and ensuring network convergence to exact solutions more reliably. Figure 21 depicts that in the best irrigation time model, TanElu achieved the highest performance with an accuracy of 97.31%. It also demonstrates that the proposed TanElu can enhance the performance of this model. The Tanh activation function, with a score of 94.15%, stands in second place, while ELU and ReLU, with accuracies of 93.87% and 93.41% respectively, and achieved nearly identical values. TanElu can generate a model that is 3% more accurate than models generated by other activation functions. The last model, ReLU, has an accuracy of 93.41%.

The Table 4 shows the overall statistical results of the ANN model in terms of using the activation functions TanElu, Tanh, ELU and ReLU for the accuracy, precession, recall and F1-score values relative to the irrigation classification model and the prediction of the best irrigation time. It is common in literature to evaluate the model to see if the system will correctly generalize [43], [44] to monitor the average performance of the proposed model.

According to Table 4, the ANN algorithm achieved the highest accuracy when using the proposed activation function

TABLE 4. Overall statistical results of ANN model using proposed activation function.

Classification/ Prediction Model	Performance Measure	ANN-TanElu	ANN-Tanh	ANN-ELU	ANN-ReLU
Irrigation classification	Accuracy %	98.24	95.54	95.73	96.38
	Precision %	92.11	91.20	88.00	93.75
	Recall %	97.77	81.01	86.03	83.80
	F1-score %	94.85	85.80	87.01	88.50
Best Time Prediction	Accuracy %	97.31	94.15	93.87	93.41
	Precision %	93.99	89.15	96.58	99.33
	Recall %	98.36	94.26	84.97	81.15
	F1-score %	96.13	91.63	90.41	89.32

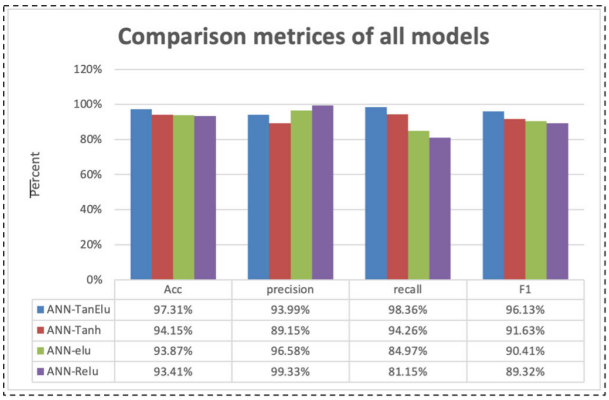


FIGURE 21. Comparison between TanElu and other activation functions.

TanElu compared to the other activation functions, as the classification need of water or not and prediction of the best irrigation time accuracy reached 98.24% and 97.31%, respectively. The hybrid activation function combining Tanh and ELU offers a balanced advantage by leveraging the strengths of both functions for classifying irrigation and predicting the best irrigation time. This combination excels in capturing intricate patterns, handling noise, and adapting to changing conditions, making it well-suited for real-time predictions. In contrast, other activation functions like ReLU, Sigmoid, and Tanh have specific strengths but might struggle with vanishing gradients, negative values, or nonlinearity. The hybrid approach's ability to simultaneously address these challenges contributes to higher accuracy in predicting optimal irrigation times and effectively classifying irrigation, underscoring its superior suitability for these tasks.

4) COMPARISON OF FORECASTING MODELS

Using machine learning regression to convert a time series problem into a supervised format is advantageous as it enables the application of established supervised algorithms, feature engineering, and structured tabular data to time-dependent sequences. This conversion facilitates the utilisation of powerful predictive models, leverages feature extraction from historical observations, and accommodates variable-length inputs by fixing input lengths. Which ultimately enhancing prediction accuracy and leveraging the strengths of established supervised learning techniques.

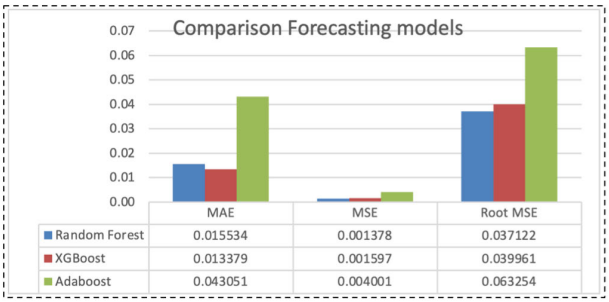


FIGURE 22. Comparison forecasting model.

Figure 22 shows the metrics of forecasting models, pecifically Random Forest, XGboost, and AdaBoost which are popular ensemble learning techniques used in machine learning regression tasks due to their ability to improve predictive accuracy, handle complex relationships, and mitigate overfitting. Each performs well in capturing complex nonlinear relationships, providing accurate predictions, and reducing prediction errors through iterative reinforcement. There are several common metrics for evaluating the performance of regression models. Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Since the target value is not balanced, the most appropriate metric for evaluating prediction reversion is MAE. In addition to its simplicity, robustness, and ease of interpretation, it is a commonly used metric, which makes it useful for evaluating the accuracy of regression models as it is easy to understand in real-world applications. The performance of all models is observed to be very close, with XGboost having a minor difference and the best model having an MAE of 0.0013379. Adaboost, on the other hand, has the lowest performance with an MAE of 0.043051. The performance of all models is so close together, which shows that it is possible to replace any of them for forecasting environmental variables.

5) COMPARISON OF THE EXECUTION TIME OF THE PROPOSED ACTIVATION FUNCTION TANELU AND OTHER FUNCTIONS FOR COLLECTED DATASET

Using execution time in the smart watering code is important because it allows for the measurement of the time it takes for a specific operation to complete. This measurement enables

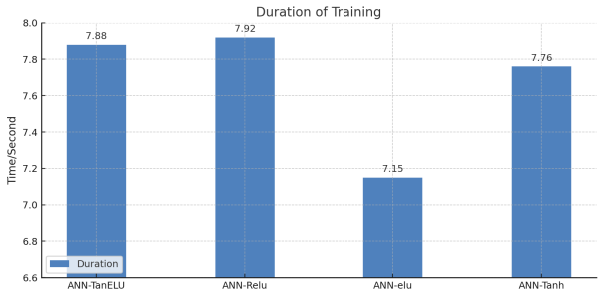


FIGURE 23. Execution time for irrigation classification.

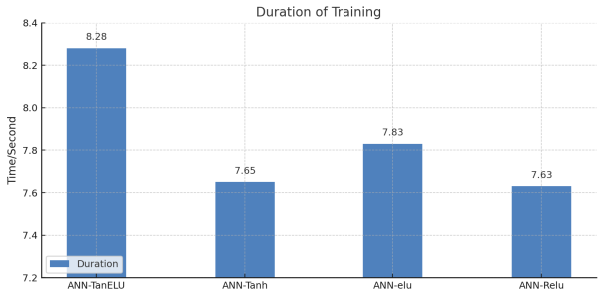


FIGURE 24. Execution time for best time prediction.

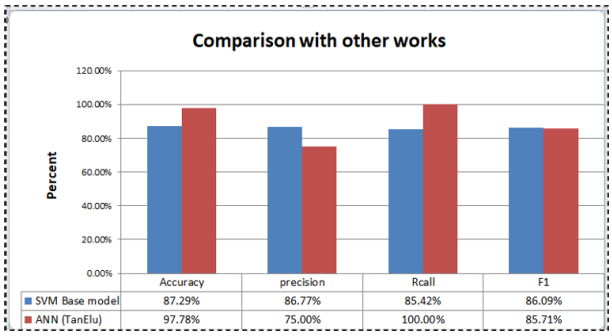


FIGURE 25. Comparison forecasting model.

the identification and improvement of the code parts that may be causing performance issues. This section compares the different activation functions with the proposed activation function in terms of model execution time. Based on Figure 23, it is evident that the proposed activation function TanElu required 7.88 seconds for training in irrigation classification. This duration is longer when compared to the Tanh and ELU, separately. However, it was slightly faster than the ReLU function. The reason is that the model must generate values for both the Tanh and ELU activation functions, then compare the outputs and select the maximum value. This leads to a slightly higher calculation complexity compared to each individual activation function, which in turn affects the training time.

On the other hand, the execution time of the best-time prediction model using TanElu's proposed activation function took 8.28 seconds, which is the longest, while ELU took 7.83 seconds. It shows that the proposed activation function does not significantly increase the modeling time by virtue of the fact that the training time for each model is comparable. As mentioned earlier, TanElu is a combination of Tanh and Elu and includes more steps of calculation; thus, the training time of this model is also increased. Execution time is shown in Figure 24.

The Table 5 demonstrates that the remaining activation functions performed better than the proposed activation function TanhElu on the collected dataset at the time of execution. Thus, TanhElu takes longer than the others. This is because the proposed activation function is generated by the combination of two exponential functions, Tanh and ELU. As can be seen in Table 5, the variance in execution time is relatively small.

From this section, it can be concluded that TanElu promotes good performance and improves classification accuracy simultaneously in the two selected datasets; as well it improves the prediction of the best irrigation time for the collected dataset. To further optimize the computational efficiency of TanElu, future work may consider function approximation techniques, pruning of redundant computations, or implementing the function using hardware-efficient operations such as lookup tables or fused operations.

6) COMPARISON RESULTS THE PROPOSED MODEL WITH EXISTING DATASET AND THE COLLECTED DATASET

Comparing the enhancement model's performance with another model from the literature on the same dataset is crucial to gauge the relative effectiveness and innovation of the approach. This comparison provides a direct and fair assessment of how the enhancement model fares against an established method under consistent conditions. By analyzing metrics such as accuracy, precision, recall, and F1 score, we can quantitatively determine if the model's enhancements lead to improved predictions, potentially revealing new insights or uncovering overlooked patterns. This comparative analysis validates the value of the contributions, substantiates the claims of improved performance, and enhances the overall understanding of the model's strengths and limitations in comparison to existing solutions. Figure 25 presents a comparison of the metrics between existing dataset and the collected dataset by using ANN model with proposed activation function in this work. Notably, the proposed model demonstrated superior performance compared to the SVM algorithm that was used in the study paper. The proposed model achieved an accuracy of 97.78% on the same data set. This accuracy is 10% higher than the accuracy stated in the study paper which was 87.29%. In addition, the accuracy achieved by the DT and NB algorithms in the core paper was 85.83% and 83.61%, respectively, both of which were lower than the accuracy of the improved ANN algorithm.

Moreover, and according to Figure 26 which shows that the accuracy of the combined data set is 98.24% which is

TABLE 5. Execution time of TanElu and other activation functions.

Classification/ Prediction Model	ANN-TanElu	ANN-Tanh	ANN-ELU	ANN-ReLU
Irrigation classification	7.76	7.15	7.92	7.88
Best Time Prediction	7.65	7.83	7.63	8.28

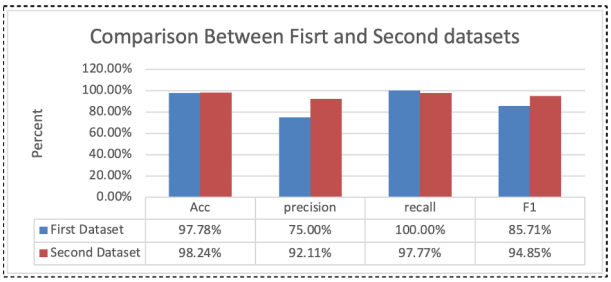


FIGURE 26. Comparison between two datasets by using ANN model with TanElu activation function.

about 1% higher than the baseline dataset when using the same proposed model in two different sets of data, while the accuracy of other metrics, recall and F1 score has a significant difference compared to the baseline model. This shows that preparing more data with high quality assists models in outperforming. The deep learning models require a large amount of data to demonstrate their capabilities, and this is easily observed in this study.

Comparing the performance of the proposed optimization model on a collected dataset with a dataset from the literature serves a crucial purpose in validating the effectiveness and generalizability of the proposed approach. The rationale behind the use of an enhanced model of ANN using proposed hybrid activation function (TanElu) is in its inherent ability to effectively deal with complex and nonlinear relationships within complex datasets. While SVMs excel in linearly separable classes, they can falter when faced with intricate or nonlinear patterns. In contrast, the neural network's architecture, coupled with the unique benefits of the hybrid activation function (TanElu), allows it to capture complex nonlinear relationships more adeptly. This choice also mitigates the need for extensive manual feature engineering, common in SVM applications. Additionally, the neural network's adaptability to various data complexities and its potential for better generalization underscore its suitability for scenarios where the data relationships are intricate and require flexible modeling, making it a more robust choice than the SVM algorithm. The irrigation process in the base paper is carried out manually, whereas in the proposed model, it is automated by sending a high-voltage signal to activate the pumps whenever watering is required. Furthermore, the base paper did not suggest the irrigation time. Therefore, this aspect can be regarded as a gap in the base paper, which has been addressed in the proposed model. The study underscores the pivotal role of activation functions in influencing the success of neural network architectures, particularly when confronted with a comprehensive real-world dataset derived from local farms. This dataset embodies real-time nonlinear relationships that challenge the network's ability to discern intricate patterns. Notably, the TanElu activation function reached to remarkable accuracy rate of 98.24% on this collected dataset. The success of TanElu is attributed to its hybrid nature, amalgamating Tanh and ELU properties. Tanh's capability to map values within the range of -1 to 1 aids in centered distribution around zero, enhancing convergence during training. Conversely, ELU mitigates vanishing gradient challenges by facilitating the use of negative values, particularly advantageous for deep networks. This hybrid activation function effectively captures the intricate complexities inherent in the dataset, outperforming single activation functions. The narrative also highlights the practical implications of the chosen activation function in real-time applications, particularly in irrigation timing models. TanElu, with an accuracy of 97.31%, outperforms other activation functions, indicating its aptitude for accurate predictions even in rapidly changing conditions. This utility is attributed to its ability to swiftly adapt to new information and sustain network convergence. The presented data portrays TanElu as 3% more accurate than models produced by alternative activation functions. On the other hand, the TanElu activation function demonstrated slightly prolonged execution times in training due to its amalgamation of Tanh and ELU, the assessment of best-time prediction models illustrated its relatively minor influence on overall modeling time. Despite this, the comparative execution time analysis indicated that the TanhElu activation function exhibited longer execution times compared to other activation functions, primarily attributed to its inherent complexity stemming from the fusion of exponential functions. This can be attributed to the computational complexity introduced by TanElu's mechanism of computing values for both Tanh and ELU activations and subsequently selecting the maximum, leading to a nuanced increase in calculation steps that influence training time.

In summation, the study vividly portrays the significance of the TanElu activation function in augmenting machine learning model performance. It substantiates the function's potency in capturing intricate data patterns, adapting to dynamic conditions, and enhancing model accuracy and stability. The study's findings underscore the critical role of activation function selection in crafting effective neural network architectures tailored to complex datasets and real-time scenarios.

V. CONCLUSION AND FUTURE WORK

The implementation of a smart irrigation system utilising IoT and machine learning presents a promising avenue for revolutionizing water management practices in agriculture. This study focused on data collection and processing using an ANN model with the proposed hybrid activation function which called TanElu. The primary objective was to accurately predict plant water requirements and optimise irrigation timing. The results indicated that the proposed model attained superior performance in irrigation classification, and the prediction of irrigation time was highly accurate. By training and evaluating the model on two different sets of data, this study has showcased its exceptional performance in irrigation classification and precise timing prediction. Furthermore, this study establishes the superiority of the proposed model over existing literature-based models, illustrating its potential impact on advancing irrigation practices. The implications of this study extend beyond academia, with potential benefits for agricultural stakeholders; farmers can achieve cost savings, increase in crop yield and quality, and reduce water waste. To fully realize these benefits, future research should explore the integration of additional IoT devices, such as weather sensors, and rigorously assess the economic viability across diverse regions and farming methods. In general, this study demonstrates that smart irrigation can play a crucial role in sustainable agriculture and contribute significantly to global efforts in water conservation. Future studies in intelligent IoT-based smart irrigation systems could focus on expanding to more crops and environments, optimizing irrigation schedules through machine learning, and improving system accessibility with user-friendly interfaces and multilingual support. Conducting surveys and field trials would ensure the systems meet farmers' needs and align with stakeholder requirements. Additionally, we recognize that the dataset used in this study is relatively small, which may limit the generalizability of the model across broader agricultural scenarios. While the model has shown promising results, future work will aim to address this limitation by collecting larger and more diverse datasets. Expanding the dataset across different seasons, soil types, and geographic locations will enhance the robustness and adaptability of the proposed model in real-world deployments and validating the system through real-world field trials to assess its performance and practicality in actual farming conditions.

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